

Automated Analyzers & POCT - Practice Exam

I. Multiple Choice Questions (MCQ)

1. In Continuous Flow Analyzers, different samples are separated by:

- a) Physical barriers | b) Air bubbles | c) Centrifugal force**

Answer: b

2. The first "Autoanalyzer" was a:

- a) Discrete Analyzer | b) Single-channel Continuous-flow | c) POCT device**

Answer: b

3. Multi-channel continuous flow analyzers can perform how many tests simultaneously?

- a) 1 or 2 | b) 6 or 12 | c) 100 or 200**

Answer: b

4. Which analyzer uses centrifugal force for mixing and reaction?

- a) Discrete Analyzer | b) Centrifugal Analyzer | c) Batch Analyzer**

Answer: b

5. Which analyzer treats each sample in a separate (discrete) cuvette?

- a) Continuous Flow | b) Discrete Analyzer | c) Sequential Analyzer**

Answer: b

6. A Batch Analyzer is characterized by:

- a) Different tests at once | b) One type of test at a time | c) Portable design**

Answer: b

7. Random Access Analyzers allow specimens to be analyzed:

- a) Alphabetically | b) Out of sequence | c) Manually**

Answer: b

8. POCT stands for:

- a) Primary Care Tool | b) Point of Care Testing | c) Patient Care Tech**

Answer: b

9. A major advantage of POCT is:

a) Large size | b) Rapid results for immediate decisions | c) High complexity

Answer: b

10. "Bed-side testing" is another name for:

a) Lab testing | b) POCT | c) Continuous flow

Answer: b

II. True or False

1. Continuous Flow Analyzers use a peristaltic pump. (True)
2. Centrifugal analyzers have low efficiency. (False - High efficiency)
3. Discrete analyzers can run multiple tests on one sample. (True)
4. POCT is performed only inside the central laboratory. (False - Near the patient)
5. Random Access Analyzers are more flexible than Batch Analyzers. (True)
6. Computers in analyzers help in storing and analyzing data. (True)
7. Discrete analyzers keep samples in a continuous tubing. (False - Separate wells)
8. POCT results aid in immediate decision-making. (True)
9. Single-channel analyzers process 40 samples per hour. (True)
10. Ancillary testing is a synonym for POCT. (True)

III. Fill in the Blanks (Complete)

In continuous flow, samples are separated by Air bubbles.

Multi-channel analyzers run multiple tests on one sample simultaneously.

Centrifugal force is used to complete the Chemical reaction.

Discrete analyzer performs tests in separate cuvettes or cells.

Batch analyzers use only one type of Reagent at a time.

POCT is designed to be simple and Portable.

The first Autoanalyzer provided about 40 samples per hour.

Detection in continuous flow measures Absorbance.

POCT is also known as Near-patient testing.

Random Access analyzers allow tests to be done in any order.

IV. Short Notes

1. Define Continuous Flow Analyzers.

Samples/reagents pass through the same path, separated by air bubbles, moved by a peristaltic pump.

2. Mention an advantage of Centrifugal Analyzers.

Simultaneous mixing and detection leading to very high efficiency.

3. What is a Random Access Analyzer?

A device that can perform any test on any sample at any time, out of sequence.

4. Define POCT.

Clinical testing performed near the patient (outside the lab) for rapid results.

5. List 3 names for POCT.

Bed-side testing, Near-patient testing, Physician office testing.

6. Why use Automation in labs?

To increase speed, ensure precision, and reduce human errors.

7. What is the role of the Peristaltic Pump?

To provide a constant, continuous flow of liquids through the system.

8. How many results can Multi-channel analyzers produce?

From 6 to 12 test results simultaneously for a single sample.

9. What is a Batch Analyzer?

An analyzer that performs only one specific type of test on a group of samples at once.

10. Compare Lab vs POCT feedback time.

Lab feedback takes 1-2 days; POCT feedback is immediate (minutes).